

Breather & Revision

What to review (pick at least one per section)

❖ Day 04/05 processes and service

`systemctl status nginx`: this command to check the status of systemd services and units on linux systems, it displays whether a service is active, inactive, failed, or disabled.

`journalctl -u nginx`: this command prints log entries from systemd journal.

❖ Day 03 process management commands, file system, Networking troubleshooting.

`free -h`: it shows the total amount of free and physical and swap memory in the system, as well as the buffer used by the kernel. ‘-h’ shows all output fields automatically scaled to the shortest three-digit units and display the units.

`mv`: this command is used to move or rename files and directories from one location to another

`traceroute`: this command used to determine the path taken by packets from the source to the destination host. It also shows the number the hops required to reach the destination.

`netstat`: this command is used to display network connections, routing tables, interface statistics, and network protocol information on linux.

`ping`: this command is a network utility that tests connectivity between your computer and another device on a network. It sends small data packets to the target and measures how long it takes to receive a response, helping you determine if the connection is working and how fast it is.

❖ Day 06-11 file skills: practice 3 quick ops

File permission issue: A script at `/home/user/backup.sh` is not executed. When you run it: `./backup.sh` you get: “permission denied”, what commands would you use to fix this?

`ls -l`: this command is used to see the permission of the file.

Add execute permission: `chmod +x /home/user/backup.sh`

verify it worked

`ls -l /home/user/back.sh`

```
ubuntu@ip-172-31-42-193:~$ cd home/ubuntu/
-bash: cd: home/ubuntu/: No such file or directory
ubuntu@ip-172-31-42-193:~$ cd /home/ubuntu
ubuntu@ip-172-31-42-193:~$ vim backup.sh
ubuntu@ip-172-31-42-193:~$ ./backup.sh
-bash: ./backup.sh: No such file or directory
ubuntu@ip-172-31-42-193:~$ ./backup.sh
-bash: ./backup.sh: Permission denied
ubuntu@ip-172-31-42-193:~$ chmod +x /home/ubuntu/ba
backup.sh  bank_heist/
ubuntu@ip-172-31-42-193:~$ chmod +x /home/ubuntu/backup.sh
ubuntu@ip-172-31-42-193:~$ ls -l /home/
berlin/      nairobi/    professor/  tokyo/      ubuntu/
ubuntu@ip-172-31-42-193:~$ ls -l /home/
berlin/      nairobi/    professor/  tokyo/      ubuntu/
ubuntu@ip-172-31-42-193:~$ ls -l /home/ubuntu/backup.sh
-rwxrwxr-x 1 ubuntu ubuntu 30 Feb 11 08:41 /home/ubuntu/backu
p.sh
```

How to view nginx logs: `tail -f /var/nginx/access.log`

How to save nginx logs to text file: `tail -f /var/nginx/access.log | tee nginx_log.txt`

```
ubuntu@ip-172-31-42-193:~$ tail -f /var/log/nginx/access.log | tee nginx_log.txt
1.38.220.202 - - [11/Feb/2026:08:54:12 +0000] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
1.38.220.202 - - [11/Feb/2026:08:54:13 +0000] "GET /favicon.ico HTTP/1.1" 404 196 "http://13.127.151.100/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
```

How to download log files to your local machine: `scp -i`

`"user_group_management.pem" ubuntu@ ec2-13-127-151-100.ap-south-1.compute.amazonaws.com:/home/ubuntu/nginx_log.txt .`

```
rohan@MSI_MINGW64 ~/downloads
$ scp -i "user_group_management.pem" ubuntu@ec2-13-127-151-100.ap-south-1.compute.amazonaws.com:/home/ubuntu/nginx_log.txt .
nginx_log.txt
rohan@MSI_MINGW64 ~/downloads
```

Write three lines into the file using redirection (`>` and `>>`): `echo "I am rohan. i am learning devOps engineering " > notes.txt && echo "I learn new concepts every day." >> notes.txt && echo " Everyday assignment help me to understand new concepts." >> notes.txt`

```
ubuntu@ip-172-31-42-193:~$ echo "I am rohan. i am learning devOps engineering" > notes.txt && echo "I learn new concepts everyday." >> notes.txt && echo "Everyday assignment help me to understand new concepts." >> notes.txt
ubuntu@ip-172-31-42-193:~$ cat notes.txt
I am rohan. i am learning devOps engineering
I learn new concepts everyday.
Everyday assignment help me to understand new concepts.
ubuntu@ip-172-31-42-193:~$
```

User_group recreate one small scenario from day-09 or day-11(create a user or change ownership) verify by id or ls -l

```
ubuntu@ip-172-31-42-193:~$ sudo useradd -m helsinki -s /usr/bin/bash
ubuntu@ip-172-31-42-193:/home$ ls
berlin helsinki nairobi professor tokyo ubuntu
ubuntu@ip-172-31-42-193:/home$ sudo passwd helsinki
New password:
Retype new password:
passwd: password updated successfully
ubuntu@ip-172-31-42-193:/home$ tail -n 5 /etc/group
planners:x:1009:
vault_team:x:1010:
tech_team:x:1011:
helsinki:x:1012:
team_arms:x:1013:
ubuntu@ip-172-31-42-193:/home$ id helsinki
uid=1005(helsinki) gid=1013(team_arms) groups=1013(team_arms)
ubuntu@ip-172-31-42-193:/home$
```

❖ Mini self-check (write short answer in day 12 revision.md)

➤ Which three commands saves you the most time right now and why

Ans: chmod +x /path, id user, sudo chgrp group name username:

These above three commands save me the most time now. Because in user and group management, these commands are key to managing the permissions; id is to verify whether the group ownership is changed or not. Chmod + /path is used to give all the execute permissions, sudo chgrp group name username is used to change the owner ship of the group.

- How do you check if a service is healthy? List the exact command 2–3-day commands you had run first?
 - Ans: Top: Top command displays a list of processes that are running in real-time along with their memory and CPU usage.
 - Ps: ps is short for 'Process Status'. It displays the current running processes. However, unlike the top command, the output generated is not in real-time.
 - kill: To stop a process in Linux, use the 'kill' command. kill command sends a signal to the process.
 - DF: This utility reports the free disk space (Hard Disk) on all the file systems.
 - Free: This command shows the free and used memory (RAM) on the Linux system.
 - Nice: it starts a new process(job) and assigns its priority (nice) value at the same time.
 - Cp: The cp command is used to copy files and directories from one location to another.
 - Host: the host command is used to perform DNS lookup. It can find ip address associated with a domain name.
- How do you safely change ownership and permission without breaking access? Give one example command?
- Ans: sudo chown helsinki:vault_team /home/helsinki/
- Sudo chmod 755 helsinki/

```
ubuntu@ip-172-31-42-193:/home$ sudo chown helsinki:vault_team /home/helsinki/
ubuntu@ip-172-31-42-193:/home$ ls -l
```

```
ubuntu@ip-172-31-42-193:/home$ sudo chmod 755 helsinki/
ubuntu@ip-172-31-42-193:/home$ ls -l
total 24
drwxr-x--- 2 berlin    berlin    4096 Feb  9 06:17 berlin
drwxr-xr-x 2 helsinki  vault_team 4096 Feb 11 10:03 helsinki
drwxr-x--- 2 nairobi   nairobi   4096 Feb  9 07:11 nairobi
drwxr-x--- 2 professor professor 4096 Feb  8 11:12 professor
drwxr-x--- 2 tokyo     tokyo     4096 Feb  8 12:34 tokyo
drwxr-x--- 7 ubuntu    ubuntu    4096 Feb 11 12:24 ubuntu
```